

A Levinson-type theorem for Schrödinger operators on a one-dimensional lattice with varying spectral multiplicity

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August 12, 2026

Abstract

In this paper, we prove a Levinson-type theorem for Schrödinger operators on the Hilbert space of vector-valued square-integrable functions over the integers. The operators that we consider are sums of a weighted discrete Laplacian and a matrix-valued potential with sufficiently fast decay. Our theorem relates the integral over the logarithmic derivative of the determinant of the scattering matrix to the number of bound states and half-bound states of the model. The scattering matrix varies in dimension, wherefore novel techniques are needed for the analysis of the integrand around points with dimensional jumps.

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1 Introduction

In this paper, we consider the following model: Given $L \in \mathbb{N}$ and an invertible normal matrix $A \in \text{GL}_L(\mathbb{C})$, we define the free Hamiltonian H_0 on $\mathcal{H} := \ell^2(\mathbb{Z}, \mathbb{C}^L)$ by

$$(H_0\phi)(n) := A^*\phi(n+1) + A\phi(n-1), \quad n \in \mathbb{Z}.$$

Further, we consider a potential

$$(V\phi)(n) := V(n)\phi(n), \quad n \in \mathbb{Z}, \phi \in \mathcal{H},$$

where $V(n) \in \mathbb{C}^{L \times L}$ are self-adjoint $(L \times L)$ -matrices satisfying the growth condition

$$\sum_{n \in \mathbb{Z}} \|V(n)\| n^2 \tau_A^{|n|} < \infty,$$

where

$$\tau_A := \eta_A + \sqrt{\eta_A^2 - 1} \quad \text{with} \quad \eta_A := \|A\| \cdot \|A^{-1}\|.$$

(This growth condition is optimal, see [BS26a, TODO].) We define the interaction Hamiltonian

$$H := H_0 + V$$

and write S for the scattering operator associated to the pair (H_0, H) . Further, we consider the the direct integral decomposition

$$S = \int_{\sigma(H_0)}^{\oplus} S^E dE,$$

where S^E is the scattering matrix at energy E .

This model has been studied in [BS26a], [BS26b], and [BS26c], where the existence of Jost solutions has been proven, the scattering matrix has been constructed, a limiting absorption principle has been proven, and the wave and scattering operator have been calculated. It has also been shown that the dimension of the scattering matrix S^E depends on the energy E and that the function $E \mapsto S^E$ exhibits nontrivial singular behavior at energies where dimensional jumps occur. Related models have recently been of interest, for example, in [APRR25], where in a similar model, the case of a potential V supported in a single point has been studied. In the case $A = \mathbf{1}_L$, where no dimensional jumps occur inside the spectrum of H_0 , which greatly simplifies the analysis of the scattering matrix, a Levinson-type theorem has been proven in [BFNS24].

In this paper, we prove a Levinson-type theorem for matrices $A \notin \mathbb{C}\mathbf{1}_L$. Concretely, we show that the integral

$$\frac{1}{2\pi i} \int_{\sigma(H_0)} \frac{\det(S^E)'}{\det(S^E)} dE$$

exists as a principal value and that its value is given by

$$L - \left(J_b + \frac{1}{2} J_h \right)$$

(Theorem 4.2.4), where $J_b \in \mathbb{N}$ is the number of bound states of the Hamiltonian H , i.e.

$$J_b := \sum_{E \in \mathbb{R}} \dim \ker(H - E),$$

and J_h the number of half-bound states, i.e.

$$J_h := \sum_{E \in \mathbb{R}} \dim \left(c(H, E) \setminus \ker(H - E) \right),$$

where $c(H, E)$ denotes the space of functions $\phi : \mathbb{Z} \rightarrow \mathbb{C}^L$ that formally satisfy the eigenvalue equation $H\phi = E\phi$ and for which additionally the limits

$$\lim_{n \rightarrow \pm\infty} \phi(n)$$

exist (Definition 2.1). The main idea of the proof is to exploit a relation between the scattering matrix S^E , which is not directly amenable to computation, and the Wronskians $W^{E,\pm}$ of two generalized eigenfunctions, where the sign $\pm$ indicates whether the generalized eigenfunctions are obtained as limits from the upper or lower half-plane (Definition 3.4, Proposition 4.2.3).

The structure of this paper is as follows:

In Section 2, we introduce generalized eigenfunctions for our model and present some of their basic properties as well as an analysis of their analytic behavior.

In Section 3, we then use these results to analyze the local behavior of the Wronskian. We use the fact that, by [BS26a], the Wronskian can locally be expanded as

$$W^{X,\sigma} = W^{E,\sigma} + \sqrt{X - E}^\sigma D^{E,\sigma} + (X - E) R^{E,\sigma}(X),$$

denoting by

$$\sqrt{\cdot}^\sigma : \mathbb{C} \setminus \sigma i(-\infty, 0) \rightarrow \mathbb{C}$$

the holomorphic square root with

$$\sqrt{[0, \infty)}^\sigma = [0, \infty).$$

In the first three subsections of Section 3, we then analyze successively the matrices $W^{E,\sigma}$, $D^{E,\sigma}$, and $R^{E,\sigma}(X)$, especially determining their kernels, which allows us to determine when $W^{X,\sigma}$ behaves in leading order like a constant, like $\sqrt{X - E}^\sigma$ or like $(X - E)$. It turns out that a square root term occurs precisely for every half-bound state (Corollary 3.1.2), while a linear term occurs for every bound state (Corollary 3.2.3). In the last subsection we then synthesize these results in a local description of the determinant of the Wronskian.

In Section 4, we exploit the relation

$$\frac{\det(S^E)'}{\det(S^E)} = \frac{\det(W^{E,-})'}{\det(W^{E,-})} - \frac{\det(W^{E,+})'}{\det(W^{E,+})}$$

to use the argument principle to prove that

$$\frac{1}{2\pi i} \int_{\sigma(H_0)} \frac{\det(S^E)'}{\det(S^E)} dE = L - \left(J_b + \frac{1}{2} J_h \right)$$

with the left-hand side being interpreted as a principal value around each of the poles of the integrand (Definition 4.1.5, Theorem 4.2.4).

2 Generalized Eigenfunctions

In this section, we introduce generalized eigenfunctions and present some of their basic properties that we will use in later sections.

Definition 2.1. For $N \in \mathbb{N}$, we write $(\mathbb{C}^{L \times N})^{\mathbb{Z}}$ for the set of all functions $\phi : \mathbb{Z} \rightarrow \mathbb{C}^{L \times N}$ and consider the operator

$$H : (\mathbb{C}^{L \times N})^{\mathbb{Z}} \rightarrow (\mathbb{C}^{L \times N})^{\mathbb{Z}}, \quad (H\phi)(n) = A^* \phi(n+1) + V(n)\phi(n) + A\phi(n-1).$$

(We are especially interested in the cases $N = 1$, i.e. $\mathbb{C}^L$ valued functions and $N = L$, i.e. square-matrix-valued functions.) Further, for $E \in \mathbb{C}$, we define the space of *generalized eigenfunctions*

$$\mathcal{G}_N(H, E) := \{\phi \in (\mathbb{C}^{L \times N})^{\mathbb{Z}} : H\phi = E\phi\}, \quad \mathcal{G}(H, E) := \mathcal{G}_1(H, E)$$

and for $\tau \in \{+, -\}$, we define the subspaces

$$c_\tau(H, E) := \left\{ \phi \in \mathcal{G}(H, E) : \lim_{n \rightarrow \tau\infty} \phi(n) \text{ exists} \right\} + \left\{ \phi \in \mathcal{G}(H, E) : \lim_{n \rightarrow \tau\infty} (-1)^n \phi(n) \text{ exists} \right\}$$

$$\ell_\tau^2(H, E) := \left\{ \phi \in \mathcal{G}(H, E) : \sum_{n \in \tau\mathbb{N}} \|\phi(n)\|^2 < \infty \right\}$$

and

$$c(H, E) := c_-(H, E) \cap c_+(H, E)$$

$$\ell^2(H, E) := \ell_-^2(H, E) \cap \ell_+^2(H, E).$$

We call a function $\phi \in \ell^2(H, E)$ a *bound state* and a function $\phi \in c(H, E) \setminus \ell^2(H, E)$ a *half-bound state*. We denote the number of linearly independent bound and half-bound states by

$$J_b^E := \dim(\ell^2(H, E)) \quad \text{and} \quad J_h^E := \dim\left(c(H, E) \setminus \ell^2(H, E)\right).$$

Remark 2.2. Functions in $\mathcal{G}(H, E)$ are called generalized eigenfunctions since

$$\mathcal{G}(H, E) \supseteq c(H, E) \supseteq \ell^2(H, E) = \mathcal{G}(H, E) \cap \ell^2(\mathbb{Z}, \mathbb{C}^L) = \ker(H - E).$$

Since every function in $\mathcal{G}(H, E)$ is uniquely determined by its values at two consecutive integers, it is easy to see that $\dim \mathcal{G}(H, E) = 2L$ (see [BS26a]).

In order to classify these generalized eigenfunctions, we will make the following simplification: In [BS26a, TODO], it has been shown that, after an appropriate unitary transformation on $\mathcal{H} = \ell^2(\mathbb{Z}, \mathbb{C}^L)$, one can, without loss of generality, assume that

$$A = A^* = \text{diag}(a_1, \dots, a_L) \quad \text{with} \quad a_1 \geq a_2 \geq \dots \geq a_L > 0.$$

We will make this assumption throughout the rest of the paper. With this simplification, we can make the following definition:

Definition 2.3. We write

$$\mathbb{C}_\pm := \{z \in \mathbb{C} : \pm \operatorname{Im}(z) > 0\}$$

and define

$$\mathfrak{a} := \sigma(2A) \cup \sigma(-2A) = \{2a_1, \dots, 2a_L\} \cup \{-2a_1, \dots, -2a_L\}.$$

Further, for $E \in \mathbb{R}$, let $P_{>}^E$ be the orthogonal projection onto the subspace

$$\operatorname{span}\{e_k : |E| > |2a_k|\},$$

denoting by e_k the unit-vector into the k -th coordinate direction. Similarly, we define $P_{<}^E$, $P_{\leq}^E$, $P_{\geq}^E$, $P_{=}^E$, and $P_{\neq}^E$, replacing $>$ in the definition with the respective sign. (Notice that $P_{=}^E \neq 0$, precisely if $E \in \mathfrak{a}$.)

Theorem 2.4. [BS26a, TODO] *Let $\sigma, \tau \in \{+, -\}$. Then there exists an open subset $\Omega^\sigma \subseteq \mathbb{C} \setminus \mathfrak{a}$ with $\mathbb{C}_\sigma \cup \mathbb{R} \setminus \mathfrak{a} \subseteq \Omega^\sigma$ such that, for every $E \in \Omega^\sigma \cup \mathfrak{a}$ there exist generalized eigenfunctions $u_\tau^{E,\sigma}, v_\tau^{E,\sigma} \in \mathcal{G}_L(H, E)$ with the following properties:*

- (a) $\mathcal{G}(H, E) = u_\tau^{E,\sigma} \mathbb{C}^L \oplus v_\tau^{E,\sigma} \mathbb{C}^L$.
- (b) For every $n \in \mathbb{Z}$, the map $\Omega^\sigma \ni E \mapsto u_\tau^{E,\sigma}(n) \in \mathbb{C}^{L \times L}$ is holomorphic (i.e. every matrix entry is holomorphic).
- (c) If $E \in \mathbb{C}_\sigma \cup \mathbb{R} \setminus [-2a_1, 2a_1]$, one has

$$\ell_\tau^2(H, E) = u_\tau^{E,\sigma} \mathbb{C}^L.$$

- (d) If $E \in \mathbb{R}$, one has

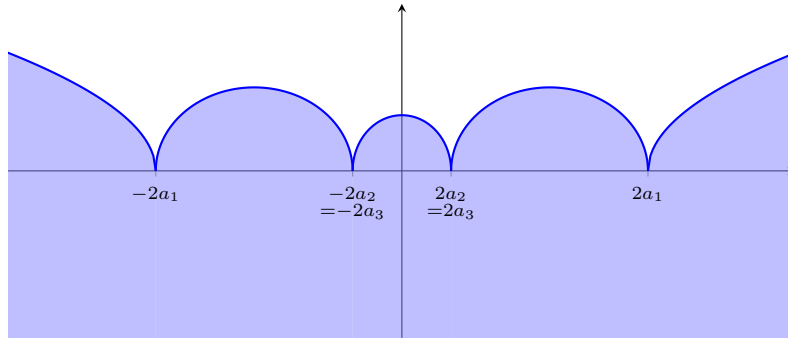
$$c_\tau(H, E) = u_\tau^{E,\sigma} P_{\leq}^E \mathbb{C}^L \quad \text{and} \quad \ell_\tau^2(H, E) = u_\tau^{E,\sigma} P_{<}^E \mathbb{C}^L.$$

- (e) If $E \in \mathbb{R}$, one has

$$u_\tau^{E,+} P_{\leq}^E = u_\tau^{E,-} P_{\leq}^E.$$

- (f) $\sup_{E \in \mathbb{C}_\sigma \cup \mathbb{R} \setminus \mathfrak{a}} \left(\limsup_{n \rightarrow \tau\infty} \left\| \frac{d}{dE} u_\tau^{E,\sigma}(n) \right\| \right) < \infty$.

Remark 2.5. Due to the symmetry of the problem, one can choose Ω^- to be the conjugate of Ω^+ (see [BS26a, TODO]).



Example of Ω^- for $L = 3$

We immediately get the following result:

Lemma 2.6. *Let $E \in \mathbb{C} \setminus [-2a_1, 2a_1]$. Then*

$$c(H, E) = \ell^2(H, E) = \ker(H - E).$$

In particular, for $E \in \mathbb{C} \setminus \mathbb{R}$, one has $J_b^E = J_h^E = 0$ and, for $E \in \mathbb{R} \setminus [-2a_1, 2a_1]$, one has

$$J_h^E = 0 \quad \text{and} \quad J_b^E = \dim \ker(H - E).$$

Proof. The inclusions

$$\ker(H - E) = \ell^2(H, E) \subseteq c(H, E)$$

are immediate. We have to show that $c(H, E) \subseteq \ker(H - E)$. So, let $\phi \in c(H, E)$. Choose $\sigma \in \{+, -\}$ such that $E \in \mathbb{C}_\sigma \cup \mathbb{R} \setminus [-2a_1, 2a_1]$. Then, by Theorem 2.4(d), there exist $\rho_\pm \in P_{\leq}^E \mathbb{C}^L$ such that

$$u_-^{E, \sigma} \rho_- = \phi = u_+^{E, \sigma} \rho_+.$$

By Theorem 2.4(b), this implies

$$\phi \in \ell_-^2(H, E) \cap \ell_+^2(H, E) = \ell^2(H, E) = \ker(H - E).$$

Thus $c(H, E) = \ell^2(H, E) = \ker(H - E)$, which implies $J_h^E = 0$ and $J_b^E = \dim \ker(H - E)$. Since H is self-adjoint, for $E \in \mathbb{C} \setminus \mathbb{R}$, one has $\ker(H - E) = \{0\}$ and therefore $J_b^E = 0$. $\square$

We will frequently use the following spaces:

Definition 2.7. For $E \in \mathbb{R}$ and $\tau \in \{+, -\}$, we define

$$\mathfrak{c}_\tau^E := \{\rho \in \mathbb{C}^L : u_\tau^{E, \sigma} \rho \in c(H, E)\} \quad \text{and} \quad \mathfrak{b}_\tau^E := \{\rho \in \mathbb{C}^L : u_\tau^{E, \sigma} \rho \in \ell^2(H, E)\}.$$

We further consider the decomposition

$$\mathbb{C}^L = \mathfrak{c}_\tau^E \oplus \mathfrak{r}_\tau^E = \mathfrak{b}_\tau^E \oplus \mathfrak{h}_\tau^E \oplus \mathfrak{r}_\tau^E$$

with

$$\mathfrak{r}_\tau^E := (\mathfrak{c}_\tau^E)^\perp \quad \text{and} \quad \mathfrak{h}_\tau^E := (\mathfrak{b}_\tau^E)^\perp \cap \mathfrak{c}_\tau^E.$$

Finally, we define the map

$$\mathcal{C}_\tau^E : \mathfrak{c}_\tau^E \rightarrow c(H, E), \quad \rho \mapsto u_\tau^{E, \sigma} \rho.$$

Remark 2.8. The definition of both $\mathfrak{c}_\tau^E$ and $\mathcal{C}_\tau^E$ is independent of $\sigma \in \{+, -\}$, since the condition $u_\tau^{E, \sigma} \rho \in c(H, E)$, by Theorem 2.4(d), implies $\rho \in P_{\leq}^E \mathbb{C}^L$, which, by Theorem 2.4(e), implies

$$u_\tau^{E, +} \rho = u_\tau^{E, -} \rho.$$

Lemma 2.9. *For $E \in \mathbb{R}$ and $\tau \in \{+, -\}$, the map $\mathcal{C}_\tau^E$ is an isomorphism with*

$$\mathcal{C}_\tau^E (\mathfrak{b}_\tau^E) = \ell^2(H, E).$$

Proof. By Theorem 2.4(a) and Remark 2.2, we have

$$L \geq \dim (u_\tau^{E, \sigma} \mathbb{C}^L) = \dim(\mathcal{G}(H, E)) - \dim (v_\tau^{E, \sigma} \mathbb{C}^L) \geq 2L - L = L$$

and therefore $\dim (u_\tau^{E, \sigma} \mathbb{C}^L) = L$, which implies the injectivity of $\mathcal{C}_\tau^E$. The surjectivity follows immediately by Theorem 2.4(d). The last equality follows by definition of $\mathfrak{b}_\tau^E$. $\square$

Since every point $E \in \mathbb{C} \setminus \mathfrak{a}$ is either contained in Ω^+ or Ω^- , we know that we can choose generalized eigenfunctions in an analytic fashion around every point except those in $\mathfrak{a}$. To better understand the analytic behavior around the points in $\mathfrak{a}$, we need the following definition:

Definition 2.10. For $\sigma \in \{+, -\}$, we consider the continuous square root

$$\sqrt{\cdot}^\sigma : \mathbb{C} \setminus \sigma i(-\infty, 0) \rightarrow \mathbb{C}$$

that is holomorphic on $\mathbb{C} \setminus \sigma i(-\infty, 0]$ and satisfies

$$\sqrt{[0, \infty)}^\sigma = [0, \infty).$$

Remark 2.11. Since both functions

$$\mathbb{C} \setminus \sigma i(-\infty, 0] \rightarrow \mathbb{C}, \quad z \mapsto \sqrt{z}^\sigma \quad \text{and} \quad \mathbb{C} \setminus \sigma i(-\infty, 0] \rightarrow \mathbb{C}, \quad z \mapsto \overline{\sqrt{z}^{-\sigma}}$$

are holomorphic and for $z \in [0, \infty)$, one has $\overline{\sqrt{z}^{-\sigma}} = \sqrt{z}^\sigma$, by the identity theorem, one has

$$\overline{\sqrt{z}^{-\sigma}} = \sqrt{z}^\sigma \quad \forall z \in \mathbb{C} \setminus \sigma i(-\infty, 0].$$

Theorem 2.12. [BS26a, TODO] *Let $E \in \mathfrak{a}$ and $\sigma, \tau \in \{+, -\}$. Then, there exists a generalized eigenfunction $d_\tau^{E, \sigma} \in \mathcal{G}_L(H, E)$ and a continuous function $r_\tau^{E, \sigma} : \Omega^\sigma \cup \mathfrak{a} \rightarrow (\mathbb{C}^{L \times L})^\mathbb{Z}$ such that, for every $X \in \Omega^\sigma \cup \mathfrak{a}$, one has*

$$u_\tau^{X, \sigma} = u_\tau^{E, \sigma} + \sqrt{X - E}^\sigma d_\tau^{E, \sigma} + (X - E)r_\tau^{E, \sigma}(X).$$

Remark 2.13. If $E \in \Omega^\sigma$, by Theorem 2.4(b), we have

$$u_\tau^{X, \sigma} = u_\tau^{E, \sigma} + (X - E)r_\tau^{E, \sigma}(X)$$

for some continuous function $r_\tau^{E, \sigma} : \Omega^\sigma \cup \mathfrak{a} \rightarrow (\mathbb{C}^{L \times L})^\mathbb{Z}$. Thus, setting $d^{E, \tau} := 0$ for $E \in \Omega^\sigma$, for every $E \in \Omega^\sigma \cup \mathfrak{a}$, we have

$$u_\tau^{X, \sigma} = u_\tau^{E, \sigma} + \sqrt{X - E}^\sigma d_\tau^{E, \sigma} + (X - E)r_\tau^{E, \sigma}(X)$$

for some continuous function $r_\tau^{E, \sigma} : \Omega^\sigma \cup \mathfrak{a} \rightarrow (\mathbb{C}^{L \times L})^\mathbb{Z}$.

3 Analysis of the Wronskian

In this section, we introduce the Wronskian. After some general analysis, we derive local expansions up to zeroth, half, and first order. The Wronskian is defined as follows:

Definition 3.1. (cf. [BS26b, TODO]) Let $N, M \in \mathbb{N}$, $\phi \in (\mathbb{C}^{L \times N})^\mathbb{Z}$, and $\psi \in (\mathbb{C}^{L \times M})^\mathbb{Z}$. We define their *Wronskian* $W(\phi, \psi) \in (\mathbb{C}^{N \times M})^\mathbb{Z}$ by

$$W(\phi, \psi)(n) := i[\phi(n+1)^* A \psi(n) - \phi(n)^* A \psi(n+1)].$$

Remark 3.2. One easily verifies that, for $N_1, N_2, M_1, M_2 \in \mathbb{N}$, matrices $B \in \mathbb{C}^{N_1 \times N_2}$, $C \in \mathbb{C}^{M_1 \times M_2}$ and functions $u : \mathbb{Z} \rightarrow \mathbb{C}^{L \times N_1}$ and $v : \mathbb{Z} \rightarrow \mathbb{C}^{L \times M_1}$, one has

$$W(uB, vC)(n) = B^* W(u, v)(n) C \quad \forall n \in \mathbb{Z}.$$

Further, it is easy to see that, for $\phi \in \mathcal{G}_N(H, E)$ and $\psi \in \mathcal{G}_M(H, E)$, the expression $W(\phi, \psi)(n)$ is independent of $n \in \mathbb{Z}$ (see [BS26b, TODO]). In this case, we usually drop the dependency on n and just write $W(\phi, \psi)$.

Remark 3.3.

We will be particularly interested in the following Wronskian:

Definition 3.4. For $\sigma \in \{+, -\}$ and $E \in \Omega^\sigma \cup \mathfrak{a}$, we set

$$W^{E, \sigma} := W \left(u_-^{\bar{E}, -\sigma}, u_+^{E, \sigma} \right).$$

Remark 3.5. By Remark 2.5 and Theorem 2.4(b), one easily sees that the map

$$\Omega^\sigma \ni E \mapsto W^{E, \sigma} \in \mathbb{C}^{L \times L}$$

is holomorphic.

We have the following result:

Proposition 3.6. [BS26b, TODO] *Let $\sigma, \tau \in \{+, -\}$ and $E \in \Omega^\sigma \cup \mathfrak{a}$. Then, the following hold:*

- (a) $W(u_\tau^{\bar{E}, -\sigma}, u_\tau^{E, \sigma}) = 0$.
- (b) $W(u_\tau^{\bar{E}, -\sigma}, v_\tau^{E, \sigma})$ is invertible.
- (c) If $E \in \mathbb{R}$, one has $-\sigma\tau W(u_\tau^{E, \sigma}, u_\tau^{E, \sigma}) \geq 0$ with $\ker W(u_\tau^{E, \sigma}, u_\tau^{E, \sigma}) = P_\leq^E \mathbb{C}^L$.
- (d) $\lim_{|E| \rightarrow \infty} E^{-1} W^{E, \sigma} = i \mathbf{1}_L$.

In this section, we want to analyze the local behavior of the function

$$\Omega^\sigma \cup \mathfrak{a} \ni E \mapsto W^{E, \sigma} \in \mathcal{M}_{L \times L}(\mathbb{C}).$$

By Remark 2.13, we have

$$u_\pm^{X, \sigma} = u_\pm^{E, \sigma} + \sqrt{X - E}^\sigma d_\pm^{E, \sigma} + (X - E) r_\pm^{E, \sigma}(X),$$

which yields

$$u_+^{X, \sigma} = u_+^{E, \sigma} + \sqrt{X - E}^\sigma d_+^{E, \sigma} + (X - E) r_+^{E, \sigma}(X)$$

and

$$\begin{aligned} u_-^{\bar{X}, -\sigma} &= u_-^{\bar{E}, -\sigma} + \sqrt{\bar{X} - \bar{E}}^{-\sigma} d_-^{\bar{E}, -\sigma} + (\bar{X} - \bar{E}) r_-^{\bar{E}, -\sigma}(\bar{X}) \\ &= u_-^{\bar{E}, -\sigma} + \overline{\sqrt{X - E}}^\sigma d_-^{\bar{E}, -\sigma} + (\bar{X} - \bar{E}) r_-^{\bar{E}, -\sigma}(\bar{X}), \end{aligned}$$

using Remark 2.11 in the last step. This yields

$$W^{X, \sigma} = W^{E, \sigma} + \sqrt{X - E}^\sigma D^{E, \sigma} + (X - E) R^{E, \sigma}(X), \quad (1)$$

where

$$D^{E,\sigma} := W\left(d_-^{\bar{E},-\sigma}, u_+^{E,\sigma}\right) + W\left(u_-^{\bar{E},-\sigma}, d_+^{E,\sigma}\right) \quad (2)$$

and

$$\begin{aligned} R^{E,\sigma}(X) := & \left[W\left(r_-^{\bar{E},-\sigma}(\bar{X}), u_+^{E,\sigma}\right) + W\left(u_-^{\bar{E},-\sigma}, r_+^{E,\sigma}(X)\right) + W\left(d_-^{\bar{E},-\sigma}, d_+^{E,\sigma}\right) \right] \\ & + \sqrt{X - E}^\sigma \left[W\left(r_-^{\bar{E},-\sigma}(\bar{X}), d_+^{E,\sigma}\right) + W\left(d_-^{\bar{E},-\sigma}, r_+^{E,\sigma}(X)\right) \right] \\ & + (X - E)W\left(r_-^{\bar{E},-\sigma}(\bar{X}), r_+^{E,\sigma}(X)\right). \end{aligned}$$

In the following three subsections, we analyze the terms $W^{E,\sigma}$, $D^{E,\sigma}$ and $R^{E,\sigma}$ respectively, in order to determine when each of them dominates the local behavior of the function $X \mapsto W^{X,\sigma}$ around E .

3.1 Zero-order approximation of the Wronskian

In this section, we will analyze when the term $W^{E,\sigma}$ is dominating the expansion (1), i.e. when $W^{E,\sigma}$ is invertible. We have the following result:

Proposition 3.1.1. *Let $\sigma \in \{+, -\}$ and $E \in \Omega^\sigma \cup \mathfrak{a}$. Then the following hold:*

- (a) *If $E \in \mathbb{C} \setminus \mathbb{R}$, then $W^{E,\sigma}$ is invertible.*
- (b) *If $E \in \mathbb{R}$, then*

$$\ker(W^{E,\sigma}) = \mathfrak{c}_+^E \quad \text{and} \quad \ker(W^{E,\sigma})^* = \mathfrak{c}_-^E.$$

Proof. Let $\rho \in \mathbb{C}^L$. Then, by Theorem 2.4(a), there exist $\alpha, \beta \in \mathbb{C}^L$ such that

$$u_+^{E,\sigma} \rho = u_-^{E,\sigma} \alpha + v_-^{E,\sigma} \beta.$$

This implies that

$$\begin{aligned} W^{E,\sigma} \rho &= W\left(u_-^{\bar{E},-\sigma}, u_+^{E,\sigma} \rho\right) = W\left(u_-^{\bar{E},-\sigma}, u_-^{E,\sigma} \alpha + v_-^{E,\sigma} \beta\right) \\ &= W\left(u_-^{\bar{E},-\sigma}, u_-^{E,\sigma}\right) \alpha + W\left(u_-^{\bar{E},-\sigma}, v_-^{E,\sigma}\right) \beta = W\left(u_-^{\bar{E},-\sigma}, v_-^{E,\sigma}\right) \beta, \end{aligned}$$

using Proposition 3.6(a) in the last step. Since $W\left(u_-^{\bar{E},-\sigma}, v_-^{E,\sigma}\right)$ is invertible by Proposition 3.6(b), we thus have $\rho \in \ker(W^{E,\sigma})$, if and only if $\beta = 0$, i.e. if and only if

$$u_+^{E,\sigma} \rho = u_-^{E,\sigma} \alpha. \quad (3)$$

- (a) For $E \in \mathbb{C} \setminus \mathbb{R}$, by Theorem 2.4(c), from Equation (3), we get

$$u_+^{E,\sigma} \rho \in \ell_+^2(H, E) \cap \ell_-^2(H, E) = \ell^2(H, E) = \{0\},$$

using Lemma 2.6 for the last equality. By Lemma 2.9, this implies $\ker(W^{E,\sigma}) = \{0\}$.

(b) For $E \in \mathbb{R}$, by Proposition 3.6(c) and Equation (3), we have

$$\begin{aligned} 0 &\leq -\sigma \left\langle \rho \middle| W(u_+^{E,\sigma}, u_+^{E,\sigma}) \rho \right\rangle = -\sigma W(u_+^{E,\sigma} \rho, u_+^{E,\sigma} \rho) \\ &= -\sigma W(u_-^{E,\sigma} \alpha, u_-^{E,\sigma} \alpha) = -\sigma \left\langle \alpha \middle| W(u_-^{E,\sigma}, u_-^{E,\sigma}) \alpha \right\rangle \leq 0, \end{aligned}$$

and therefore

$$0 = \left\langle \rho \middle| W(u_+^{E,\sigma}, u_+^{E,\sigma}) \rho \right\rangle = \left\langle \alpha \middle| W(u_-^{E,\sigma}, u_-^{E,\sigma}) \alpha \right\rangle.$$

By Proposition 3.6(c), this implies $\rho, \alpha \in P_{\leq}^E \mathbb{C}^L$, which, by Theorem 2.4(d) and Equation (3) implies

$$u_+^{E,\sigma} \rho = u_-^{E,\sigma} \alpha \in c_+(H, E) \cap c_-(H, E) = c(H, E)$$

and therefore $\rho \in \mathfrak{c}_+^E$. Conversely, for $\rho \in \mathfrak{c}_+^E$, by Theorem 2.4(d), there exists $\alpha \in P_{\leq}^E \mathbb{C}^L$ such that Equation (3) holds. This shows $\ker(W^{E,\sigma}) = \mathfrak{c}_+^E$. Using that

$$(W^{E,\sigma})^* = \left(W(u_-^{\bar{E}, -\sigma}, u_+^{E,\sigma}) \right)^* = W(u_+^{E,\sigma}, u_-^{\bar{E}, -\sigma}),$$

the equation

$$\ker(W^{E,\sigma})^* = \mathfrak{c}_-^E$$

is shown the same way by exchanging the roles of $(+, \sigma, E)$ and $(-, -\sigma, \bar{E})$. $\square$

In the following, given two subspaces $X, Y \subseteq \mathbb{C}^L$ and $M \in \mathbb{C}^{L \times L}$, we define

$$M|_X^Y : X \rightarrow Y, \quad x \mapsto P_Y Mx,$$

where P_Y is the orthogonal projection onto Y . With this notation, as an immediate consequence of Proposition 3.1.1, we get the following:

Corollary 3.1.2. *Let $\sigma \in \{+, -\}$ and $E \in \mathbb{R}$. Then, one has*

$$W^{E,\sigma}|_{\mathfrak{c}_+^E}^{\mathbb{C}^L} = 0, \quad W^{E,\sigma}|_{\mathfrak{c}_-^E}^{\mathbb{C}^L} = 0,$$

and the matrix

$$W^{E,\sigma}|_{\mathfrak{r}_+^E}^{\mathfrak{r}_-^E}$$

is invertible.

3.2 Half-order approximation of the Wronskian

In Corollary 3.1.2, we have seen that, if $\rho \in \mathfrak{r}_+^E$, one has $W^{E,\sigma} \rho \neq 0$ and therefore, by (1), we have

$$W^{X,\sigma} \rho \approx W^{E,\sigma} \rho \quad \text{for} \quad |X - E| \ll 1.$$

For $\rho \in \mathfrak{c}_+^E$ however, we have $W^{E,\sigma} \rho = 0$ and therefore

$$W^{X,\sigma} \rho = \sqrt{X - E}^\sigma D^{E,\sigma} \rho + (X - E) R^{E,\sigma}(X) \rho.$$

Similar statements hold for the adjoint. In these cases, we thus have to analyze the term $D^{E,\sigma}$. We start with the following result:

Lemma 3.2.1. *Let $\sigma, \tau \in \{+, -\}$, $E \in \mathbb{R}$ and $\rho \in P_{<}^E \mathbb{C}^L$. Then, the following hold:*

(a) *For every $n \in \mathbb{Z}$, the function*

$$\mathbb{R} \ni X \mapsto u_{\tau}^{X, \sigma}(n) \rho \in \mathbb{C}^L$$

is analytic at E .

(b) $\limsup_{n \rightarrow \tau \infty} \left\| \frac{d}{dE} u_{\tau}^{E, \sigma}(n) \rho \right\| < \infty$

(c) $d_{\tau}^{E, \sigma} \rho = 0$.

Proof. (a) For $E \in \mathbb{R} \setminus \mathfrak{a}$, the statement follows by Theorem 2.4(b). We can thus assume that $E \in \mathfrak{a}$. Choose $\varepsilon > 0$ such that $U_{\varepsilon}(E) \cap \mathfrak{a} = \{E\}$. Then, for every $X \in U_{\varepsilon}(E) \cap \mathbb{R}$, one has $P_{<}^X \rho = P_{<}^E \rho$ and therefore $\rho \in P_{<}^X \mathbb{C}^L$. Therefore, by Theorem 2.4(b),(e) and Theorem 2.12, for every $n \in \mathbb{Z}$, the function

$$U_{\varepsilon}(E) \ni X \mapsto \begin{cases} u_{\tau}^{X, +}(n) \rho & \text{if } X \in \mathbb{C}_+ \\ u_{\tau}^{X, +}(n) \rho = u_{\tau}^{X, -} \rho & \text{if } X \in \mathbb{R} \\ u_{\tau}^{X, -}(n) \rho & \text{if } X \in \mathbb{C}_- \end{cases}$$

is continuous on $U_{\varepsilon}(E)$ and holomorphic on $U_{\varepsilon}(E) \setminus \{E\}$. By Riemann's theorem on removable singularities the function is then holomorphic on all of $U_{\varepsilon}(E)$.

(b) This follows immediately by Theorem 2.4(f) and the continuity of the derivative of a holomorphic function.

(c) This follows from the fact that an analytic function cannot have $\sqrt{\cdot}^{\sigma}$ -terms in its power expansion. $\square$

Proposition 3.2.2. *Let $\sigma \in \{+, -\}$, $E \in \mathbb{R}$ and $\rho_{\pm} \in \mathbb{C}^L$. Then, the following hold:*

(a) *One has*

$$W \left(u_{\tau}^{\bar{E}, -\sigma}, d_{\tau}^{E, \sigma} \right) = \tau i \sqrt{\frac{E}{2}} P_{\equiv}^E.$$

(b) *If $\rho_+ \in \mathfrak{c}_+^E$ and $\rho_- \in \mathfrak{c}_-^E$, one has*

$$\langle \rho_- | D^{E, \sigma} \rho_+ \rangle = i \sqrt{\frac{E}{2}} \left[\langle P_{\equiv}^E \rho_- | (\mathcal{C}_{-}^E)^{-1} \mathcal{C}_{+}^E \rho_+ \rangle + \langle (\mathcal{C}_{+}^E)^{-1} \mathcal{C}_{-}^E \rho_- | P_{\equiv}^E \rho_+ \rangle \right].$$

(c) *If $\rho_+ \in \mathfrak{b}_+^E$ or $\rho_- \in \mathfrak{b}_-^E$, one has $\langle \rho_- | D^{E, \sigma} \rho_+ \rangle = 0$.*

Proof. (a) Let $\tilde{u}_{\tau}^{\bar{E}, -\sigma}$, $\tilde{v}_{\tau}^{E, \sigma}$, and $\tilde{d}_{\tau}^{E, \sigma}$ denote the functions $u_{\tau}^{\bar{E}, -\sigma}$, $v_{\tau}^{E, \sigma}$, and $d_{\tau}^{E, \sigma}$ in the case of an absent potential $V = 0$. By [BS26a, TODO] and [BS26b, TODO], we have

$$\tilde{d}_{\tau}^{E, \sigma} = -\tau \frac{\sqrt{2E}^{\sigma}}{E} \tilde{v}_{\tau}^{E, \sigma} P_{\equiv}^E \quad \text{and} \quad W \left(\tilde{u}_{\tau}^{\bar{E}, -\sigma}, \tilde{v}_{\tau}^{E, \sigma} \right) P_{\equiv}^E = -i \frac{E}{2} P_{\equiv}^E.$$

By [BS26b, TODO], this yields

$$\begin{aligned} W \left(u_{\tau}^{\bar{E}, -\sigma}, d_{\tau}^{E, \sigma} \right) &= W \left(\tilde{u}_{\tau}^{\bar{E}, -\sigma}, \tilde{d}_{\tau}^{E, \sigma} \right) = W \left(\tilde{u}_{\tau}^{\bar{E}, -\sigma}, -\tau \frac{\sqrt{2E}^{\sigma}}{E} \tilde{v}_{\tau}^{E, \sigma} P_{\equiv}^E \right) \\ &= -\tau \frac{\sqrt{2E}^{\sigma}}{E} W \left(\tilde{u}_{\tau}^{\bar{E}, -\sigma}, \tilde{v}_{\tau}^{E, \sigma} \right) P_{\equiv}^E = -\tau \frac{\sqrt{2E}^{\sigma}}{E} \left[-i \frac{E}{2} P_{\equiv}^E \right] = \tau i \sqrt{\frac{E}{2}} P_{\equiv}^E. \end{aligned}$$

(b) Let $\tau \in \{+, -\}$ and $\rho_\tau \in \mathfrak{c}_\tau^E$. Using that

$$u_{-\tau}^{\bar{E}, -\sigma} \rho_{-\tau} = \mathcal{C}_{-\tau}^E \rho_{-\tau} = u_\tau^{\bar{E}, -\sigma} (\mathcal{C}_\tau^E)^{-1} \mathcal{C}_{-\tau}^E \rho_{-\tau},$$

by (a), we have

$$\begin{aligned} \langle \rho_{-\tau} | W(u_{-\tau}^{\bar{E}, -\sigma}, d_\tau^{E, \sigma}) \rho_\tau \rangle &= W(u_{-\tau}^{\bar{E}, -\sigma} \rho_{-\tau}, d_\tau^{E, \sigma} \rho_\tau) \\ &= W(u_\tau^{\bar{E}, -\sigma} (\mathcal{C}_\tau^E)^{-1} \mathcal{C}_{-\tau}^E \rho_{-\tau}, d_\tau^{E, \sigma} \rho_\tau) \\ &= \langle (\mathcal{C}_\tau^E)^{-1} \mathcal{C}_{-\tau}^E \rho_{-\tau} | W(u_\tau^{\bar{E}, -\sigma}, d_\tau^{E, \sigma}) \rho_\tau \rangle \\ &= \left\langle (\mathcal{C}_\tau^E)^{-1} \mathcal{C}_{-\tau}^E \rho_{-\tau} \left| \tau i \sqrt{\frac{E}{2}} P_\tau^E \rho_\tau \right. \right\rangle \\ &= \tau i \sqrt{\frac{E}{2}} \langle (\mathcal{C}_\tau^E)^{-1} \mathcal{C}_{-\tau}^E \rho_{-\tau} | P_\tau^E \rho_\tau \rangle. \end{aligned}$$

Setting $\tau = +$, we get

$$\langle \rho_- | W(u_-^{\bar{E}, -\sigma}, d_+^{E, \sigma}) \rho_+ \rangle = i \sqrt{\frac{E}{2}} \langle (\mathcal{C}_+^E)^{-1} \mathcal{C}_-^E \rho_- | P_+^E \rho_+ \rangle \quad (4)$$

and, setting $\tau = -$, we have

$$\begin{aligned} \langle \rho_- | W(d_-^{\bar{E}, -\sigma}, u_+^{E, \sigma}) \rho_+ \rangle &= \langle W(u_+^{E, \sigma}, d_-^{\bar{E}, -\sigma}) \rho_- | \rho_+ \rangle \\ &= \overline{\langle \rho_+ | W(u_+^{E, \sigma}, d_-^{\bar{E}, -\sigma}) \rho_- \rangle} \\ &= -i \sqrt{\frac{E}{2}} \langle (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho_+ | P_-^E \rho_- \rangle \\ &= i \sqrt{\frac{E}{2}} \langle P_-^E \rho_- | (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho_+ \rangle, \end{aligned}$$

using Remark 2.11 in the last step. The statement then follows immediately by Equation (2).

(c) For $\rho_+ \in \mathfrak{b}_+^E$, by Equation (2), Lemma 3.2.1(c), and Equation (4), we get

$$\begin{aligned} \langle \rho_- | D^{E, \sigma} \rho_+ \rangle &= \langle \rho_- | W(d_-^{\bar{E}, -\sigma}, u_+^{E, \sigma}) + W(u_-^{\bar{E}, -\sigma}, d_+^{E, \sigma}) \rho_+ \rangle \\ &= i \sqrt{\frac{E}{2}} \langle (\mathcal{C}_+^E)^{-1} \mathcal{C}_-^E \rho_- | P_+^E \rho_+ \rangle = 0, \end{aligned}$$

using in the last step that $P_+^E \rho_+ = 0$ by Theorem 2.4(d). Similarly, for $\rho_- \in \mathfrak{b}_-^E$, one has

$$\langle \rho_- | D^{E, \sigma} \rho_+ \rangle = \sqrt{\frac{E}{2}} \langle P_-^E \rho_- | (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho_+ \rangle = 0. \quad \square$$

Corollary 3.2.3. *Let $\sigma \in \{+, -\}$ and $E \in \mathbb{R}$. Then, one has*

$$D^{E,\sigma} \Big|_{\mathfrak{h}_+^E}^{\mathbb{C}^L} = 0, \quad D^{E,\sigma} \Big|_{\mathbb{C}^L}^{\mathfrak{h}_+^E} = 0,$$

and the matrix

$$D^{E,\sigma} \Big|_{\mathfrak{h}_+^E}^{\mathfrak{h}_+^E}$$

is invertible.

Proof. The first two equalities follow immediately by Proposition 3.2.2(c). To show that $D^{E,\sigma} \Big|_{\mathfrak{h}_+^E}^{\mathfrak{h}_+^E}$ is invertible, let $\rho \in \ker \left(D^{E,\sigma} \Big|_{\mathfrak{h}_+^E}^{\mathfrak{h}_+^E} \right)$. Then, by Proposition 3.2.2(b), we have

$$\begin{aligned} 0 &= \langle (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho | D^{E,\sigma} \rho \rangle = i \sqrt{\frac{E}{2}}^\sigma \left[\langle P_-^E (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho | (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho \rangle + \langle \rho | P_-^E \rho \rangle \right] \\ &= i \sqrt{\frac{E}{2}}^\sigma \left[\| P_-^E (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho \|^2 + \| P_-^E \rho \|^2 \right]. \end{aligned}$$

This implies either $E = 0$ or $P_-^E \rho = 0 = P_-^E (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho$. However $P_-^0 = 0$, so in any case we have $P_-^E \rho = 0 = P_-^E (\mathcal{C}_-^E)^{-1} \mathcal{C}_+^E \rho$. By Theorem 2.4(d), this implies $\rho \in \mathfrak{h}_+^E$, but by assumption, we have $\rho \in \mathfrak{h}_+^E$, so $\rho \in \mathfrak{h}_+^E \cap \mathfrak{h}_+^E = \{0\}$. This shows that $\rho = 0$ and therefore $D^{E,\sigma} \Big|_{\mathfrak{h}_+^E}^{\mathfrak{h}_+^E}$ is invertible. $\square$

3.3 First-order approximation of the Wronskian

We now have seen that, for $\rho \in \mathfrak{r}_+^E$, one has

$$W^{X,\sigma} \rho \approx W^{E,\sigma} \rho \quad \text{for} \quad |X - E| \ll 1$$

and, for $\rho \in \mathfrak{h}_+^E$, one has $W^{E,\sigma} \rho = 0$ and $D^{E,\sigma} \rho \neq 0$, so

$$W^{X,\sigma} \rho \approx \sqrt{X - E}^\sigma D^{E,\sigma} \rho \quad \text{for} \quad |X - E| \ll 1.$$

For $\rho \in \mathfrak{h}_+^E$ however, by Equation (2), Corollary 3.1.2 and Corollary 3.2.3, one has $W^{E,\sigma} \rho = 0$ and $D^{E,\sigma} \rho = 0$ and therefore

$$W^{X,\sigma} \rho = (X - E) R^{E,\sigma}(X) \rho,$$

with similar statements holding for the adjoint. In these cases, we thus have to analyze the term $R^{E,\sigma}(E)$ to determine the local behavior of $X \mapsto W^{X,\sigma} \rho$. The ideas in this subsection are adaptations of ideas found in [BFNS24, Sec. 5]. We start with the following result, using that, by Lemma 3.2.1(a) the function $E \mapsto u_\tau^{E,\sigma} \rho$ is differentiable:

Proposition 3.3.1. *Let $\sigma, \tau \in \{+, -\}$, $E \in \mathbb{R}$, $u \in \mathcal{G}(H, E)$ and $\rho \in P_{<}^E \mathbb{C}^L$. Then*

$$W \left(u, \frac{d}{dE} u_\tau^{E,\sigma} \rho \right) (n-1) - W \left(u, \frac{d}{dE} u_\tau^{E,\sigma} \rho \right) (n) = i \langle u(n) | u_\tau^{E,\sigma}(n) \rho \rangle, \quad n \in \mathbb{Z}.$$

Proof. Let $n \in \mathbb{Z}$. By Theorem 2.4, we have

$$Au_\tau^{E,\sigma}(n+1) + Au_\tau^{E,\sigma}(n-1) + V(n)u_\tau^{E,\sigma}(n) = Eu_\tau^{E,\sigma}(n).$$

Multiplying with $u(n)^*$ from the left and with ρ from the right and taking derivatives, we get

$$\begin{aligned} u(n)^* A \frac{d}{dE} u_\tau^{E,\sigma}(n+1) \rho + u(n)^* A \frac{d}{dE} u_\tau^{E,\sigma}(n-1) \rho + u(n)^* V(n) \frac{d}{dE} u_\tau^{E,\sigma}(n) \rho \\ = u(n)^* u_\tau^{E,\sigma}(n) \rho + u(n)^* E \frac{d}{dE} u_\tau^{E,\sigma}(n) \rho. \end{aligned} \quad (5)$$

On the other hand, we have

$$Au(n+1) + Au(n-1) + V(n)u(n) = Eu(n).$$

Taking adjoints and multiplying with $\frac{d}{dE} u_\tau^{E,\sigma}(n) \rho$ from the right leads

$$\begin{aligned} u(n+1)^* A \frac{d}{dE} u_\tau^{E,\sigma}(n) \rho + u(n-1)^* A \frac{d}{dE} u_\tau^{E,\sigma}(n) \rho + u(n)^* V(n) \frac{d}{dE} u_\tau^{E,\sigma}(n) \rho \\ = u(n)^* E \frac{d}{dE} u_\tau^{E,\sigma}(n) \rho. \end{aligned} \quad (6)$$

Now, subtracting Equation (6) from Equation (5) and multiplying by i , we get

$$-W \left(u, \frac{d}{dE} u_\tau^{E,\sigma} \rho \right) (n) + W \left(u, \frac{d}{dE} u_\tau^{E,\sigma} \rho \right) (n-1) = iu(n)^* u_\tau^{E,\sigma}(n) \rho. \quad \square$$

Proposition 3.3.2. *Let $E \in \mathbb{R}$, $\sigma \in \{+, -\}$ and $\rho_\pm \in P_{<}^E \mathbb{C}^L$. Then*

$$\langle \rho_- | R^{E,\sigma}(E) \rho_+ \rangle = i \langle \mathcal{C}_-^E \rho_- | \mathcal{C}_+^E \rho_+ \rangle_{\ell^2}.$$

Proof. Let $n \in \mathbb{Z}$. By Proposition 3.3.1, for every $l > n$, we have

$$\begin{aligned} W \left(\mathcal{C}_-^E \rho_-, \frac{d}{dE} u_+^{E,\sigma} \rho_+ \right) (n) \\ = W \left(\mathcal{C}_-^E \rho_-, \frac{d}{dE} u_+^{E,\sigma} \rho_+ \right) (l) + \sum_{m=n+1}^l i \langle (\mathcal{C}_-^E \rho_-)(m) | u_+^{E,\sigma}(m) \rho_+ \rangle \\ \xrightarrow{l \rightarrow \infty} \sum_{m=n+1}^{\infty} i \langle (\mathcal{C}_-^E \rho_-)(m) | u_+^{E,\sigma}(m) \rho_+ \rangle = i \sum_{m=n+1}^{\infty} \langle (\mathcal{C}_-^E \rho_-)(m) | (\mathcal{C}_+^E \rho_+)(m) \rangle, \end{aligned}$$

using that $\mathcal{C}_-^E \rho_- \in \ell^2(H, E)$, we have $\lim_{l \rightarrow \infty} (\mathcal{C}_-^E \rho_-)(l) = 0$ and thus, by Lemma 3.2.1(b), we have

$$\lim_{l \rightarrow \infty} W \left(\mathcal{C}_-^E \rho_-, \frac{d}{dE} u_+^{E,\sigma} \rho_+ \right) (l) = 0.$$

Thus

$$W \left(\mathcal{C}_-^E \rho_-, \frac{d}{dE} u_+^{E,\sigma} \rho_+ \right) (n) = i \sum_{m=n+1}^{\infty} \langle (\mathcal{C}_-^E \rho_-)(m) | (\mathcal{C}_+^E \rho_+)(m) \rangle.$$

Taking adjoints, one analogously sees that

$$W \left(\frac{d}{dE} u_{-}^{\bar{E}, -\sigma} \rho_{-}, \mathcal{C}_{+}^E \rho_{+} \right) (n) = i \sum_{m=-\infty}^n \langle (\mathcal{C}_{-}^E \rho_{-})(m) | (\mathcal{C}_{+}^E \rho_{+})(m) \rangle.$$

By Equation (1), we then have

$$\begin{aligned} \langle \rho_{-} | R^{E, \sigma}(E) \rho_{+} \rangle &= \frac{d}{dE} \langle \rho_{-} | W^{E, \sigma} \rho_{+} \rangle = \frac{d}{dE} W \left(u_{-}^{\bar{E}, -\sigma} \rho_{-}, u_{+}^{E, \sigma} \rho_{+} \right) (n) \\ &= W \left(\frac{d}{dE} u_{-}^{\bar{E}, -\sigma} \rho_{-}, u_{+}^{E, \sigma} \rho_{+} \right) (n) + W \left(u_{-}^{\bar{E}, -\sigma} \rho_{-}, \frac{d}{dE} u_{+}^{E, \sigma} \rho_{+} \right) (n) \\ &= i \sum_{m=-\infty}^{\infty} \langle (\mathcal{C}_{-}^E \rho_{-})(m) | (\mathcal{C}_{+}^E \rho_{+})(m) \rangle = i \langle \mathcal{C}_{-}^E \rho_{-} | \mathcal{C}_{+}^E \rho_{+} \rangle_{\ell^2}. \end{aligned} \quad \square$$

Corollary 3.3.3. *Let $E \in \mathbb{R}$ and $\sigma \in \{+, -\}$. Then the operator*

$$R^{E, \sigma}(E) \big|_{\mathfrak{b}_{+}^E}^{\mathfrak{b}_{-}^E}$$

is invertible.

Proof. Let $\rho \in \ker \left(R^{E, \sigma}(E) \big|_{\mathfrak{b}_{+}^E}^{\mathfrak{b}_{-}^E} \right)$. Then, by Proposition 3.3.2, we have

$$0 = \langle (\mathcal{C}_{-}^E)^{-1} \mathcal{C}_{+}^E \rho | R^{E, \sigma}(E) \rho \rangle = i \langle \mathcal{C}_{+}^E \rho | \mathcal{C}_{+}^E \rho \rangle_{\ell^2} = i \|\mathcal{C}_{+}^E \rho\|_{\ell^2}^2$$

and therefore $\mathcal{C}_{+}^E \rho = 0$, which, by Lemma 2.9, implies $\rho = 0$. $\square$

3.4 Determinant of the Wronskian

In this subsection, we synthesize our results of the previous subsections by analyzing the local behavior of the determinant $X \mapsto \det(W^{X, \sigma})$.

Theorem 3.4.1. *Let $\sigma \in \{+, -\}$ and $E \in \mathbb{R}$. There exists a continuous function*

$$F^{E, \sigma} : \Omega^{\sigma} \cup \mathfrak{a} \rightarrow \mathbb{C},$$

that is holomorphic on Ω^{σ} with $F^{E, \sigma}(E) \neq 0$, such that

$$\det(W^{X, \sigma}) = \left(\sqrt{X - E} \right)^{J_h^E + 2J_b^E} F^{E, \sigma}(X) \quad \forall X \in \Omega^{\sigma} \cup \mathfrak{a}.$$

Proof. Let $V_{\pm}$ be a change-of-basis-matrix, that maps the first $L - J_h^E - J_b^E$ standard base vectors to an orthonormal basis of $\mathfrak{r}_{\pm}^E$, the next J_h^E standard base vectors to an orthonormal basis of $\mathfrak{h}_{\pm}^E$, and the last J_b^E standard base vectors to an orthonormal basis of $\mathfrak{b}_{\pm}^E$. Then

$$V_{-}^{*} W^{X, \sigma} V_{+} = \begin{pmatrix} \begin{bmatrix} W^{X, \sigma} | \mathfrak{r}_{+}^E \\ W^{X, \sigma} | \mathfrak{h}_{+}^E \\ W^{X, \sigma} | \mathfrak{b}_{+}^E \end{bmatrix} & \begin{bmatrix} W^{X, \sigma} | \mathfrak{r}_{-}^E \\ W^{X, \sigma} | \mathfrak{h}_{-}^E \\ W^{X, \sigma} | \mathfrak{b}_{-}^E \end{bmatrix} & \begin{bmatrix} W^{X, \sigma} | \mathfrak{r}_{+}^E \\ W^{X, \sigma} | \mathfrak{h}_{+}^E \\ W^{X, \sigma} | \mathfrak{b}_{+}^E \end{bmatrix} \\ \begin{bmatrix} W^{X, \sigma} | \mathfrak{r}_{+}^E \\ W^{X, \sigma} | \mathfrak{h}_{+}^E \\ W^{X, \sigma} | \mathfrak{b}_{+}^E \end{bmatrix} & \begin{bmatrix} W^{X, \sigma} | \mathfrak{r}_{-}^E \\ W^{X, \sigma} | \mathfrak{h}_{-}^E \\ W^{X, \sigma} | \mathfrak{b}_{-}^E \end{bmatrix} & \begin{bmatrix} W^{X, \sigma} | \mathfrak{r}_{-}^E \\ W^{X, \sigma} | \mathfrak{h}_{-}^E \\ W^{X, \sigma} | \mathfrak{b}_{-}^E \end{bmatrix} \end{pmatrix}.$$

Then, by Corollary 3.1.2 and Corollary 3.2.3, we have

$$V_-^* W^{X,\sigma} V_+ = \begin{pmatrix} g_{rr}(X) & \sqrt{X-E}^\sigma g_{rh}(X) & (X-E)g_{rb}(X) \\ \sqrt{X-E}^\sigma g_{hr}(X) & \sqrt{X-E}^\sigma g_{hh}(X) & (X-E)g_{hb}(X) \\ (X-E)g_{br}(X) & (X-E)g_{bh}(X) & (X-E)g_{bb}(X) \end{pmatrix}$$

with the matrix-valued functions

$$\begin{aligned} g_{rr}(X) &:= W^{X,\sigma} \Big|_{\mathfrak{r}_+^E}, & g_{rb}(X) &:= R^{E,\sigma}(X) \Big|_{\mathfrak{b}_+^E}, & g_{br}(X) &:= R^{E,\sigma}(X) \Big|_{\mathfrak{r}_+^E}, \\ g_{bb}(X) &:= R^{E,\sigma}(X) \Big|_{\mathfrak{b}_+^E}, & g_{hb}(X) &:= R^{E,\sigma}(X) \Big|_{\mathfrak{b}_+^E}, & g_{bh}(X) &:= R^{E,\sigma}(X) \Big|_{\mathfrak{b}_+^E}, \end{aligned}$$

and

$$\begin{aligned} g_{rh}(X) &:= D^{E,\sigma} \Big|_{\mathfrak{h}_+^E} + \sqrt{X-E}^\sigma R^{E,\sigma}(X) \Big|_{\mathfrak{h}_+^E} \\ g_{hr}(X) &:= D^{E,\sigma} \Big|_{\mathfrak{r}_+^E} + \sqrt{X-E}^\sigma R^{E,\sigma}(X) \Big|_{\mathfrak{r}_+^E} \\ g_{hh}(X) &:= D^{E,\sigma} \Big|_{\mathfrak{h}_+^E} + \sqrt{X-E}^\sigma R^{E,\sigma}(X) \Big|_{\mathfrak{h}_+^E}. \end{aligned}$$

By construction, all these functions are continuous on $\Omega^\sigma \cup \mathfrak{a}$ and holomorphic on Ω^σ . We now define

$$f^{E,\sigma} : \Omega^\sigma \cup \mathfrak{a} \rightarrow \mathbb{C}^{L \times L}, \quad X \mapsto \begin{pmatrix} g_{rr}(X) & g_{rh}(X) & g_{rb}(X) \\ \sqrt{X-E}^\sigma g_{hr}(X) & g_{hh}(X) & g_{hb}(X) \\ (X-E)g_{br}(X) & \sqrt{X-E}^\sigma g_{bh}(X) & g_{bb}(X) \end{pmatrix}.$$

Then, for $X \in \Omega^\sigma \cup \mathfrak{a}$, we have

$$W^{X,\sigma} = (V_-^*)^{-1} f^{E,\sigma}(X) \begin{pmatrix} \mathbf{1}_{J_u^E} & 0 & 0 \\ 0 & \sqrt{X-E}^\sigma \mathbf{1}_{J_h^E} & 0 \\ 0 & 0 & (X-E)\mathbf{1}_{J_b^E} \end{pmatrix} (V_+)^{-1},$$

which yields

$$\begin{aligned} \det(W^{X,\sigma}) &= \overline{\det(V_-)}^{-1} \det(f^{E,\sigma}(X)) \left(\sqrt{X-E}^\sigma \right)^{J_h^E} (X-E)^{J_b^E} \det(V_+)^{-1} \\ &= \overline{\det(V_-)}^{-1} \det(V_+)^{-1} \det(f^{E,\sigma}(X)) \left(\sqrt{X-E}^\sigma \right)^{J_h^E + 2J_b^E}. \end{aligned}$$

We now notice that, by Corollary 3.1.2, we have

$$\det(g_{rr}(E)) = \det\left(W^{E,\sigma} \Big|_{\mathfrak{r}_+^E}\right) \neq 0,$$

by Corollary 3.2.3, we have

$$\det(g_{hh}(E)) = \det\left(D^{E,\sigma} \Big|_{\mathfrak{h}_+^E}\right) \neq 0$$

and by Corollary 3.3.3, we have

$$\det(g_{bb}(E)) = \det\left(R^{E,\sigma}(E)|_{\mathfrak{b}_+^E}\right) \neq 0.$$

This yields

$$\begin{aligned} \det(f^{E,\sigma}(E)) &= \det\begin{pmatrix} g_{rr}(E) & g_{rh}(E) & g_{rb}(E) \\ 0 & g_{hh}(E) & g_{hb}(E) \\ 0 & 0 & g_{bb}(E) \end{pmatrix} \\ &= \det(g_{rr}(E)) \cdot \det(g_{hh}(E)) \cdot \det(g_{bb}(E)) \neq 0. \end{aligned}$$

The statement then follows by defining

$$F^{E,\sigma} : \Omega^\sigma \cup \mathfrak{a} \rightarrow \mathbb{C}, \quad X \mapsto \overline{\det(V_-)}^{-1} \det(V_+)^{-1} \det(f^{E,\sigma}(X)). \quad \square$$

Corollary 3.4.2. *Let $\sigma \in \{+, -\}$ and $E \in \mathbb{R}$. Then*

$$\frac{\frac{d}{dX} \det(W^{X,\sigma})}{\det(W^{X,\sigma})} = \left(J_b^E + \frac{1}{2}J_h^E\right) \frac{1}{X-E} + \frac{\frac{d}{dX} F^{E,\sigma}(X)}{F^{E,\sigma}(X)} \quad \forall X \in \Omega^\sigma \cup \mathfrak{a}.$$

Proof. Using that

$$1 = \frac{d}{dX}(X-E) = \frac{d}{dX}(\sqrt{X-E}^\sigma)^2 = 2\sqrt{X-E}^\sigma \frac{d}{dX}\sqrt{X-E}^\sigma,$$

i.e.

$$\frac{d}{dX}\sqrt{X-E}^\sigma = \frac{1}{2\sqrt{X-E}^\sigma},$$

we have

$$\begin{aligned} \frac{d}{dX}(\sqrt{X-E}^\sigma)^{J_h^E+2J_b^E} &= (J_h^E + 2J_b^E) (\sqrt{X-E}^\sigma)^{J_h^E+2J_b^E-1} \frac{d}{dX}\sqrt{X-E}^\sigma \\ &= (J_h^E + 2J_b^E) (\sqrt{X-E}^\sigma)^{J_h^E+2J_b^E-1} \frac{1}{2\sqrt{X-E}^\sigma} \\ &= \left(J_b^E + \frac{1}{2}J_h^E\right) (\sqrt{X-E}^\sigma)^{J_h^E+2J_b^E-2} \end{aligned}$$

and therefore

$$\begin{aligned} \frac{\frac{d}{dX}(\sqrt{X-E}^\sigma)^{J_h^E+2J_b^E}}{(\sqrt{X-E}^\sigma)^{J_h^E+2J_b^E}} &= \frac{(J_b^E + \frac{1}{2}J_h^E) (\sqrt{X-E}^\sigma)^{J_h^E+2J_b^E-2}}{(\sqrt{X-E}^\sigma)^{J_h^E+2J_b^E}} \\ &= \left(J_b^E + \frac{1}{2}J_h^E\right) (\sqrt{X-E}^\sigma)^{-2} \\ &= \left(J_b^E + \frac{1}{2}J_h^E\right) \frac{1}{X-E}. \end{aligned}$$

Then, by Theorem 3.4.1, due to the additivity of the logarithmic derivative, we get

$$\begin{aligned} \frac{\frac{d}{dX} \det(W^{X,\sigma})}{\det(W^{X,\sigma})} &= \frac{\frac{d}{dX} \left(\sqrt{X-E}^\sigma \right)^{J_h^E + 2J_b^E}}{\left(\sqrt{X-E}^\sigma \right)^{J_h^E + 2J_b^E}} + \frac{\frac{d}{dX} F^{E,\sigma}(X)}{F^{E,\sigma}(X)}. \\ &= \left(J_b^E + \frac{1}{2} J_h^E \right) \frac{1}{X-E} + \frac{\frac{d}{dX} F^{E,\sigma}(X)}{F^{E,\sigma}(X)}. \end{aligned} \quad \square$$

One can use Theorem 3.4.1 to characterize the following set:

Definition 3.4.3. We define

$$\mathcal{N} := \{E \in \mathbb{C} : J_b^E \neq 0\} \cup \{E \in \mathbb{C} : J_h^E \neq 0\}.$$

Proposition 3.4.4. *The following hold:*

- (a) For $\sigma \in \{+, -\}$, one has $\mathcal{N} = \{E \in \Omega^\sigma : \det(W^{E,\sigma}) = 0\}$.
- (b) $\mathcal{N}$ is a finite subset of $\mathbb{R}$.

Proof. (a) By Lemma 2.6 and Proposition 3.1.1(a), we have

$$\mathcal{N} \subseteq \mathbb{R} \quad \text{and} \quad \{E \in \Omega^\sigma : \det(W^{E,\sigma}) = 0\} \subseteq \mathbb{R}.$$

Further, for $E \in \mathbb{R}$, by Theorem 3.4.1, we have

$$\det(W^{X,\sigma}) = \left(\sqrt{X-E}^\sigma \right)^{J_h^E + 2J_b^E} F^{E,\sigma}(X), \quad X \in \Omega^\sigma \cup \mathfrak{a} \quad (7)$$

with $F^{E,\sigma}$ being continuous and satisfying $F^{E,\sigma}(E) \neq 0$. This shows that one has $\det(W^{E,\sigma}) = 0$, if and only if

$$\left(\sqrt{E-E}^\sigma \right)^{J_h^E + 2J_b^E} = 0,$$

which is equivalent to $J_h^E + 2J_b^E > 0$, i.e. to $J_b^E \neq 0$ or $J_h^E \neq 0$.

- (b) Since $F^{E,\sigma}(E) \neq 0$ and $F^{E,\sigma}$ is continuous, for every $E \in \mathbb{R}$, there exists $\varepsilon^E > 0$, such that

$$F^{E,\sigma}(X) \neq 0 \quad \forall X \in U_{\varepsilon^E}(E).$$

Since

$$\left(\sqrt{X-E}^\sigma \right)^{J_h^E + 2J_b^E} \neq 0 \quad \forall X \in U_{\varepsilon^E}(E) \setminus \{E\},$$

by Equation (7), we have

$$\det(W^{X,\sigma}) \neq 0 \quad \forall X \in U_{\varepsilon^E}(E) \setminus \{E\},$$

which shows that the $\mathcal{N}$ is a discrete subset of $\mathbb{R}$.

Further, for $E \notin [-2a_1, 2a_1]$, by Lemma 2.6, one has $J_h^E = 0$. On the other hand, for $E \notin \sigma(H)$, one has $\ker(H-E) = \{0\}$ and therefore $J_b^E = 0$. Thus, $\mathcal{N}$ is bounded as a subset of the bounded set $[-2a_1, 2a_1] \cup \sigma(H)$. The statement then follows since bounded discrete subsets of $\mathbb{R}$ are finite by the Bolzano–Weierstrass theorem. $\square$

4 A Levinson Theorem

In this section, we will use the results of Section 3 to prove a Levinson-type theorem for our model. We will do so by integrating the logarithmic derivative of the Wronskian along certain paths in the complex plane. We will then use the Schur Complement Formula to relate the logarithmic derivative of the Wronskian to the logarithmic derivative of the scattering matrix.

4.1 Path integrals

In order to compute path integrals of the logarithmic derivative of the Wronskian, we need the following result:

Proposition 4.1.1. (cf. [BFNS24, Lem. 27]) *Let $\Omega \subseteq \mathbb{C}$, $z \in \Omega$, and $f : \Omega \rightarrow \mathbb{C}$ be continuous with $f(z) \neq 0$. Further, let*

$$\Omega' := \{\xi \in \Omega^\circ : f \text{ is differentiable at } \xi\}.$$

Then, for any piecewise differentiable curves

$$\gamma_\varepsilon : [a_\varepsilon, b_\varepsilon] \rightarrow \Omega', \quad \varepsilon > 0$$

with

$$\lim_{\varepsilon \downarrow 0} \left(\sup_{t \in [a_\varepsilon, b_\varepsilon]} |\gamma_\varepsilon(t) - z| \right) = 0, \quad (8)$$

one has

$$\lim_{\varepsilon \downarrow 0} \int_{\gamma_\varepsilon} \frac{f'(\xi)}{f(\xi)} d\xi = 0.$$

Proof. Since replacing f by $f(z)^{-1}f$ does not change the value of $\frac{f'}{f}$, we can without loss of generality assume that $f(z) = 1$. Since f is continuous, there exists $r > 0$ such that

$$f(U_r(z) \cap \Omega) \subseteq U_1(1).$$

Then, by Equation (8), there exists $\varepsilon_0 > 0$, such that for every $\varepsilon \in (0, \varepsilon_0)$, one has

$$\gamma_\varepsilon([a_\varepsilon, b_\varepsilon]) \subseteq U_r(z)$$

and therefore

$$f(\gamma_\varepsilon([a_\varepsilon, b_\varepsilon])) \subseteq U_1(1) \subseteq \mathbb{C} \setminus (-\infty, 0]. \quad (9)$$

Now, let

$$\log : \mathbb{C} \setminus (-\infty, 0] \rightarrow \mathbb{C}$$

by the main branch of the complex logarithm. Then, by (9), for every $\varepsilon \in (0, \varepsilon_0)$ and $\xi \in \gamma_\varepsilon([a_\varepsilon, b_\varepsilon])$, one has

$$\frac{f'(\xi)}{f(\xi)} = (\log \circ f)'(\xi).$$

This implies

$$\begin{aligned} \int_{\gamma_\varepsilon} \frac{f'(\xi)}{f(\xi)} d\xi &= \int_{\gamma_\varepsilon} (\log \circ f)'(\xi) d\xi = (\log \circ f)(\gamma_\varepsilon(b_\varepsilon)) - (\log \circ f)(\gamma_\varepsilon(a_\varepsilon)) \\ &\xrightarrow{\varepsilon \downarrow 0} (\log \circ f)(z) - (\log \circ f)(z) = 0, \end{aligned}$$

using Equation (8) in the penultimate step. $\square$

Corollary 4.1.2. *Let $\Omega \subseteq \mathbb{C}$ be unbounded and $g : \Omega \rightarrow \mathbb{C}$ be continuous such that the limit*

$$g(\infty) := \lim_{\substack{\xi \in \Omega \\ |\xi| \rightarrow \infty}} g(\xi)$$

exists and is non-zero. Further, let

$$\Omega' := \{\xi \in \Omega^\circ : g \text{ is differentiable in } \xi\}.$$

Then, for any piecewise differentiable curves

$$\eta_R : [a_R, b_R] \rightarrow \Omega', \quad R > 0$$

with

$$\lim_{R \rightarrow \infty} \left(\inf_{t \in [a_R, b_R]} |\eta_R(t)| \right) = \infty,$$

one has

$$\lim_{R \rightarrow \infty} \int_{\eta_R} \frac{g'(\xi)}{g(\xi)} d\xi = 0.$$

Proof. The statement follows by applying Proposition 4.1.1 to the function

$$f : \left\{ \frac{1}{w} : w \in \Omega \setminus \{0\} \right\} \cup \{0\} \rightarrow \mathbb{C}, \quad \xi \mapsto \begin{cases} g\left(\frac{1}{\xi}\right) & \text{if } \xi \neq 0 \\ g(\infty) & \text{if } \xi = 0, \end{cases}$$

the point $z = 0$ and the curves $\gamma_\varepsilon = \frac{1}{\eta_\varepsilon}$, $\varepsilon > 0$. $\square$

We are especially interested in the following curves:

Definition 4.1.3. Let $\sigma \in \{+, -\}$. For $R > 0$ and $E \in \mathbb{C}$, we consider the path

$$\gamma_R^{E, \sigma} : [0, \pi] \rightarrow \mathbb{C}, \quad t \mapsto E + Re^{\sigma it}.$$

Lemma 4.1.4. *Let $E \in \mathbb{R}$. Then, the following hold:*

- (a) *For $\sigma \in \{+, -\}$, one has $\lim_{\varepsilon \downarrow 0} \int_{\gamma_\varepsilon^{E, \sigma}} \frac{\det(W^{X, \sigma})'}{\det(W^{X, \sigma})} dX = \sigma \left(J_b^E + \frac{1}{2} J_h^E \right) \pi i$.*
- (b) *For $\sigma \in \{+, -\}$, one has $\lim_{R \rightarrow \infty} \int_{\gamma_R^{0, \sigma}} \frac{\det(W^{X, \sigma})'}{\det(W^{X, \sigma})} dX = \sigma L \pi i$.*
- (c) *If $E \in \mathbb{R} \setminus \mathfrak{a}$, one has $\lim_{\varepsilon \downarrow 0} \int_{(E-\varepsilon, E+\varepsilon)} \frac{\det(W^{X, -})'}{\det(W^{X, -})} - \frac{\det(W^{X, +})'}{\det(W^{X, +})} dX = 0$.*

Proof. (a) By Corollary 3.4.2, we have

$$\begin{aligned} \int_{\gamma_\varepsilon^{E,\sigma}} \frac{\det(W^{X,\sigma})'}{\det(W^{X,\sigma})} dX &= \int_{\gamma_\varepsilon^{E,\sigma}} \left[\left(J_b^E + \frac{1}{2} J_h^E \right) \frac{1}{X-E} + \frac{\frac{d}{dX} F^{E,\sigma}(X)}{F^{E,\sigma}(X)} \right] dX \\ &= \sigma \left(J_b^E + \frac{1}{2} J_h^E \right) \pi i + \int_{\gamma_\varepsilon^{E,\sigma}} \frac{\frac{d}{dX} F^{E,\sigma}(X)}{F^{E,\sigma}(X)} dX \\ &\xrightarrow{\varepsilon \downarrow 0} \sigma \left(J_b^E + \frac{1}{2} J_h^E \right) \pi i, \end{aligned}$$

using Proposition 4.1.1 in the last step.

(b) By Proposition 3.6(d), we have

$$\lim_{|X| \rightarrow \infty} X^{-1} W^{X,\sigma} = i \mathbf{1}_L.$$

By Corollary 4.1.2, this yields

$$\begin{aligned} 0 &= \lim_{R \rightarrow \infty} \int_{\gamma_R^{0,\sigma}} \frac{\det(X^{-1} W^{X,\sigma})'}{\det(X^{-1} W^{X,\sigma})} dX = \lim_{R \rightarrow \infty} \int_{\gamma_R^{0,\sigma}} \frac{\det(W^{X,\sigma})'}{\det(W^{X,\sigma})} - \frac{\det(X \mathbf{1}_L)'}{\det(X \mathbf{1}_L)} dX \\ &= \lim_{R \rightarrow \infty} \int_{\gamma_R^{0,\sigma}} \frac{\det(W^{X,\sigma})'}{\det(W^{X,\sigma})} dX - \sigma L \pi i, \end{aligned}$$

using in the last step that

$$\int_{\gamma_R^{0,\sigma}} \frac{\det(X \mathbf{1}_L)'}{\det(X \mathbf{1}_L)} dX = \int_{\gamma_R^{0,\sigma}} \frac{L X^{L-1}}{X^L} dX = L \int_{\gamma_R^{0,\sigma}} \frac{1}{X} dX = \sigma L \pi i.$$

(c) By Corollary 3.4.2, we have

$$\begin{aligned} &\int_{(E-\varepsilon, E+\varepsilon)} \frac{\det(W^{X,-})'}{\det(W^{X,-})} - \frac{\det(W^{X,+})'}{\det(W^{X,+})} dX \\ &= \int_{(E-\varepsilon, E+\varepsilon)} \frac{\frac{d}{dX} F^{E,-}(X)}{F^{E,-}(X)} - \frac{\frac{d}{dX} F^{E,+}(X)}{F^{E,+}(X)} dX \xrightarrow{\varepsilon \downarrow 0} 0, \end{aligned}$$

using Proposition 4.1.1 in the last step. Notice that the conditions of Proposition 4.1.1 are fulfilled precisely if $E \notin \mathfrak{a}$. $\square$

In order to prove our main theorem, we need the following definitions:

Definition 4.1.5. For a subset $M \subseteq \mathbb{C}$ and $\varepsilon > 0$, we define

$$M^{[\varepsilon]} := \{m \in M : \text{dist}(m, \mathfrak{a}) \geq \varepsilon\}$$

and

$$M^{\{\varepsilon\}} := \{m \in M : \text{dist}(m, \mathcal{N} \cup \mathfrak{a}) \geq \varepsilon\}.$$

Definition 4.1.6. For $x, y \in \mathbb{C}$, we define the path

$$L_{x,y} : [0, 1] \rightarrow \mathbb{C}, \quad t \mapsto x + t(y - x).$$

Further, for any paths $\gamma_1 : [a, b] \rightarrow \mathbb{C}$ and $\gamma_2 : [c, d] \rightarrow \mathbb{C}$ with $\gamma_1(b) = \gamma_2(c)$, we define their concatenation

$$\gamma_1 \hat{\diamond} \gamma_2 : [a, b + d - c] \rightarrow \mathbb{C}, \quad t \mapsto \begin{cases} \gamma_1(t) & \text{if } t \in [a, b] \\ \gamma_2(t - b + c) & \text{if } t \in (b, b + d - c] \end{cases}$$

and for any path $\gamma : [a, b] \rightarrow \mathbb{C}$, we define the inverse path

$$\gamma^\vee : [-b, -a] \rightarrow \mathbb{C}, \quad t \mapsto \gamma(-t).$$

Proposition 4.1.7. *We have*

$$\lim_{\varepsilon \downarrow 0} \int_{[-2a_1, 2a_1]^{[\varepsilon]}} \frac{\det(W^{X,+})'}{\det(W^{X,+})} - \frac{\det(W^{X,-})'}{\det(W^{X,-})} dX = 2\pi i \left(J_b + \frac{1}{2} J_h - L \right),$$

where

$$J_b := \sum_{E \in \mathcal{N}} J_b^E \quad \text{and} \quad J_h := \sum_{E \in \mathcal{N}} J_h^E.$$

Proof. We set

$$I := \lim_{\varepsilon \downarrow 0} \int_{[-2a_1, 2a_1]^{[\varepsilon]}} f_+(X) - f_-(X) dX, \quad \text{where} \quad f_\pm(X) := \frac{\det(W^{X,\pm})'}{\det(W^{X,\pm})}$$

By Theorem 2.4(e) and Definition 3.4, for $X \in \mathbb{R} \setminus [-2a_1, 2a_1]$, one has

$$W^{X,+} = W(u_-^{\bar{X},-}, u_+^{X,+}) = W(u_-^{\bar{X},+}, u_+^{X,-}) = W^{X,-}$$

and therefore $f_+(X) - f_-(X) = 0$. This implies that, for $R \geq 2a_1$, one has

$$I = \lim_{\varepsilon \downarrow 0} \int_{[-R, R]^{[\varepsilon]}} f_+(X) - f_-(X) dX. \quad (10)$$

By Proposition 3.4.4, the set $\mathcal{N} \cup \mathfrak{a}$ is finite. Therefore, there exists $r > 0$ such that

$$\forall x, y \in \mathcal{N} \cup \mathfrak{a} : x \neq y \Rightarrow U_r(x) \cap U_r(y) = \emptyset.$$

From now on, we assume that

$$0 < \varepsilon < r \quad \text{and} \quad R > \sup_{E \in \mathcal{N} \cup \mathfrak{a}} |E| + r.$$

Then

$$[-R, R]^{[\varepsilon]} = [-R, R]^{\{\varepsilon\}} \cup \bigcup_{E \in \mathcal{N} \setminus \mathfrak{a}} (E - \varepsilon, E + \varepsilon),$$

so, by Equation (10) and Lemma 4.1.4(c), we have

$$I = \lim_{\varepsilon \downarrow 0} \int_{[-R, R]^{\{\varepsilon\}}} f_+(X) - f_-(X) dX. \quad (11)$$

Now, let

$$\mathcal{N} \cup \mathfrak{a} = \{E_1, \dots, E_n\} \quad \text{and} \quad E_1 < E_2 < \dots < E_n.$$

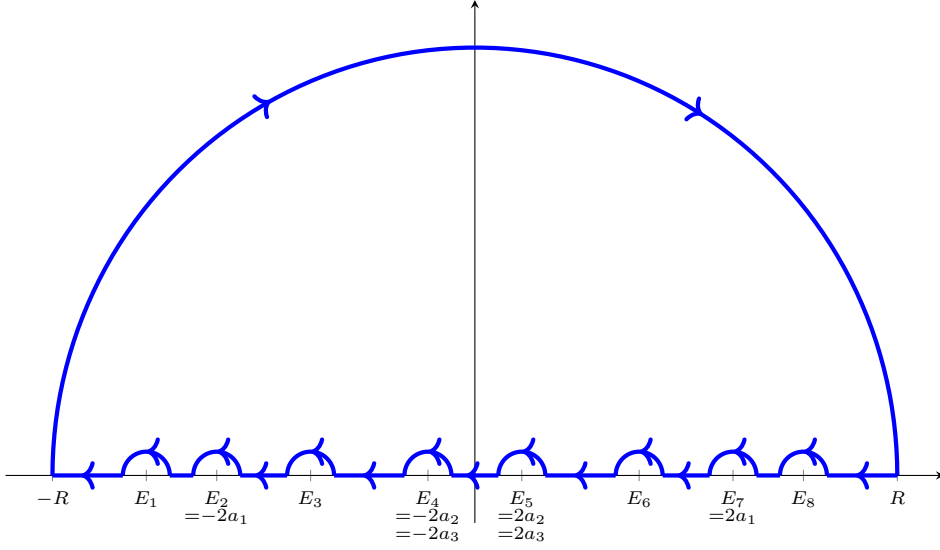
Then

$$[-R, R]^{\{\varepsilon\}} = [-R, E_1 + \varepsilon] \cup \left(\bigcup_{j=2}^n [E_{j-1} + \varepsilon, E_j - \varepsilon] \right) \cup [E_n + \varepsilon, R].$$

Now, we notice that, for $\sigma \in \{+, -\}$, the paths

$$\Gamma_{\varepsilon, R}^{\sigma} := L_{-R, E_1 - \varepsilon} \diamond \left(\gamma_{\varepsilon}^{E_1, \sigma} \right)^{\vee} \diamond L_{E_1 + \varepsilon, E_2 - \varepsilon} \diamond \dots \diamond L_{E_{n-1} + \varepsilon, E_n - \varepsilon} \diamond \left(\gamma_{\varepsilon}^{E_n, \sigma} \right)^{\vee} \diamond L_{E_n + \varepsilon, R} \diamond \gamma_R^{0, \sigma}$$

are closed and their image is contained in Ω^{σ} .



Example of $\Gamma_{\varepsilon, R}^+$ for $L = 3$ and four eigenvalues (E_1, E_3, E_6, E_8) outside of $\mathfrak{a} = \{E_2, E_4, E_5, E_7\}$

Further, by Proposition 3.4.4(a), the function $X \mapsto \det(W^{X, \sigma})$ has no zeros on or inside of $\Gamma_{\varepsilon, R}^{\sigma}$ and therefore, by the argument principle, one has

$$0 = \int_{\Gamma_{\varepsilon, R}^{\sigma}} f_{\sigma}(X) dX = \int_{[-R, R]^{\{\varepsilon\}}} f_{\sigma}(X) dX - \left[\sum_{j=1}^n \int_{\gamma_{\varepsilon}^{E_j, \sigma}} f_{\sigma}(X) dX \right] + \int_{\gamma_R^{0, \sigma}} f_{\sigma}(X) dX.$$

Using Lemma 4.1.4(a), this yields

$$\begin{aligned}
\int_{[-R,R]\{\varepsilon\}} f_\sigma(X) dX &= \left[\sum_{j=1}^n \int_{\gamma_\varepsilon^{E_j, \sigma}} f_\sigma(X) dX \right] - \int_{\gamma_R^{0, \sigma}} f_\sigma(X) dX \\
&\xrightarrow{\varepsilon \downarrow 0} \left[\sum_{j=1}^n \sigma \left(J_b^{E_j} + \frac{1}{2} J_h^{E_j} \right) \pi i \right] - \int_{\gamma_R^{0, \sigma}} f_\sigma(X) dX \\
&= \sigma \pi i \left(J_b + \frac{1}{2} J_h \right) - \int_{\gamma_R^{0, \sigma}} f_\sigma(X) dX.
\end{aligned}$$

Thus, by Equation (11), for every $R \geq \sup_{E \in \mathcal{N} \cup \mathfrak{a}} |E| + r$, one has

$$\begin{aligned}
I &= \left[+\pi i \left(J_b + \frac{1}{2} J_h \right) - \int_{\gamma_R^{0, +}} f_+(X) dX \right] - \left[-\pi i \left(J_b + \frac{1}{2} J_h \right) - \int_{\gamma_R^{0, -}} f_-(X) dX \right] \\
&= 2\pi i \left(J_b + \frac{1}{2} J_h \right) + \int_{\gamma_R^{0, -}} \frac{\det(W^{X, -})'}{\det(W^{X, -})} dX - \int_{\gamma_R^{0, +}} \frac{\det(W^{X, +})'}{\det(W^{X, +})} dX \\
&\xrightarrow{R \rightarrow \infty} 2\pi i \left(J_b + \frac{1}{2} J_h \right) + [-L\pi i] - [+L\pi i] = 2\pi i \left(J_b + \frac{1}{2} J_h - L \right),
\end{aligned}$$

using Lemma 4.1.4(b) in the penultimate step. $\square$

4.2 Relating scattering matrix and Wronskian

We will now link the above results to the scattering matrix by using the Schur Complement Formula:

Lemma 4.2.1. (Schur Complement Formula) *Let $N, M \in \mathbb{N}$, $a \in \mathbb{C}^{N \times N}$, $b \in \mathbb{C}^{N \times M}$, $c \in \mathbb{C}^{M \times N}$ and $d \in \mathbb{C}^{M \times M}$. Then the following hold:*

(a) *If a is invertible, one has*

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \det(a) \det(d - ca^{-1}b).$$

(b) *If d is invertible, one has*

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \det(d) \det(a - bd^{-1}c).$$

(c) *If d and $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ are invertible, one has*

$$\det \left(\begin{pmatrix} \mathbf{1}_N & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}_N \\ 0 \end{pmatrix} \right) = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \det(d).$$

(d) If a and d are invertible and $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is unitary, one has

$$\det \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) = \det(a) \overline{\det(d)}^{-1}.$$

Proof. (a) This follows immediately from the formula

$$\begin{pmatrix} \mathbf{1}_N & 0 \\ -ca^{-1} & \mathbf{1}_M \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & d - ca^{-1}b \end{pmatrix}$$

by taking determinants on both sides.

(b) This follows immediately from the formula

$$\begin{pmatrix} \mathbf{1}_N & -bd^{-1} \\ 0 & \mathbf{1}_M \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a - bd^{-1}c & 0 \\ c & d \end{pmatrix}$$

by taking determinants on both sides.

(c) By (b), one has

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \det(d) \det(a - bd^{-1}c) = \det(d) \det \left((\mathbf{1}_N \ 0) \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}_N \\ 0 \end{pmatrix} \right)^{-1}.$$

(d) We have

$$\begin{pmatrix} \mathbf{1}_N & 0 \\ 0 & \mathbf{1}_M \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^* \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^*a + c^*c & a^*b + c^*d \\ b^*a + d^*c & b^*b + d^*d \end{pmatrix},$$

so especially

$$\mathbf{1}_M = b^*b + d^*d \quad \text{and} \quad 0 = b^*a + d^*c.$$

The latter is equivalent to

$$ca^{-1} = -(d^{-1})^* b^*.$$

By (a), we then get

$$\begin{aligned} \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} &= \det(a) \det(d - ca^{-1}b) = \det(a) \det(d + (d^{-1})^* b^* b) \\ &= \det(a) \det((d^{-1})^*) \det(d^* d + b^* b) \\ &= \det(a) \overline{\det(d)}^{-1} \det(\mathbf{1}_M) = \det(a) \overline{\det(d)}^{-1}. \end{aligned} \quad \square$$

We will use the following result about the scattering matrix:

Proposition 4.2.2. (cf. [BS26b, TODO], [BS26c, TODO]) *Let $E \in [-2a_1, 2a_1] \setminus \mathfrak{a}$ with $\det(W^{E,\pm}) \neq 0$ and let S^E be the scattering matrix at energy E . Then, S^E is a unitary $(2N \times 2N)$ -matrix, where $N := \dim(P_{>}^E \mathbb{C}^L)$. Further, writing*

$$S^E := \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{with} \quad a, b, c, d \in \mathbb{C}^{N \times N},$$

one has

$$a = - \begin{pmatrix} m^* & 0 \end{pmatrix} (W^{E,+})^{-1} \begin{pmatrix} m \\ 0 \end{pmatrix} \quad \text{and} \quad d = \begin{pmatrix} m^* & 0 \end{pmatrix} ((W^{E,-})^*)^{-1} \begin{pmatrix} m \\ 0 \end{pmatrix}$$

for some invertible matrix $m \in \mathbb{C}^{N \times N}$.

Proposition 4.2.3. *Let $E \in [-2a_1, 2a_1] \setminus \mathfrak{a}$ with $\det(W^{E,\pm}) \neq 0$ and let S^E be the scattering matrix at energy E . Then*

$$\det(S^E) = (-1)^{\dim(P_{>}^E \mathbb{C}^L)} \frac{\det(W^{E,-})}{\det(W^{E,+})}.$$

Proof. We set $N := \dim(P_{>}^E \mathbb{C}^L)$ and let $a, b, c, d, m \in \mathbb{C}^{N \times N}$ as in Proposition 4.2.2. Then, by Lemma 4.2.1(d), we get

$$\begin{aligned} \det(S^E) &= \det(a) \overline{\det(d)}^{-1} \\ &= \det \left(- \begin{pmatrix} m^* & 0 \end{pmatrix} (W^{E,+})^{-1} \begin{pmatrix} m \\ 0 \end{pmatrix} \right) \overline{\det \left(\begin{pmatrix} m^* & 0 \end{pmatrix} ((W^{E,-})^*)^{-1} \begin{pmatrix} m \\ 0 \end{pmatrix} \right)}^{-1} \\ &= \frac{\det \left(- \begin{pmatrix} m^* & 0 \end{pmatrix} (W^{E,+})^{-1} \begin{pmatrix} m \\ 0 \end{pmatrix} \right)}{\det \left(\begin{pmatrix} m^* & 0 \end{pmatrix} (W^{E,-})^{-1} \begin{pmatrix} m \\ 0 \end{pmatrix} \right)} \\ &= \frac{\det(-\mathbf{1}_N) \det(m^*) \det \left(\begin{pmatrix} \mathbf{1}_N & 0 \end{pmatrix} (W^{E,+})^{-1} \begin{pmatrix} \mathbf{1}_N \\ 0 \end{pmatrix} \right) \det(m)}{\det(m^*) \det \left(\begin{pmatrix} \mathbf{1}_N & 0 \end{pmatrix} (W^{E,-})^{-1} \begin{pmatrix} \mathbf{1}_N \\ 0 \end{pmatrix} \right) \det(m)} \\ &= (-1)^N \frac{\det \left(\begin{pmatrix} \mathbf{1}_N & 0 \end{pmatrix} (W^{E,+})^{-1} \begin{pmatrix} \mathbf{1}_N \\ 0 \end{pmatrix} \right)}{\det \left(\begin{pmatrix} \mathbf{1}_N & 0 \end{pmatrix} (W^{E,-})^{-1} \begin{pmatrix} \mathbf{1}_N \\ 0 \end{pmatrix} \right)}. \end{aligned}$$

By Lemma 4.2.1(c), this yields

$$\begin{aligned} \det(S^E) &= (-1)^N \frac{\det(W^{E,+})^{-1} \det \left(\begin{pmatrix} 0 & \mathbf{1}_{L-N} \end{pmatrix} W^{E,+} \begin{pmatrix} 0 \\ \mathbf{1}_{L-N} \end{pmatrix} \right)}{\det(W^{E,-})^{-1} \det \left(\begin{pmatrix} 0 & \mathbf{1}_{L-N} \end{pmatrix} W^{E,-} \begin{pmatrix} 0 \\ \mathbf{1}_{L-N} \end{pmatrix} \right)} \\ &= (-1)^N \frac{\det(W^{E,-})}{\det(W^{E,+})} \cdot \frac{\det \left(\begin{pmatrix} 0 & \mathbf{1}_{L-N} \end{pmatrix} W^{E,+} \begin{pmatrix} 0 \\ \mathbf{1}_{L-N} \end{pmatrix} \right)}{\det \left(\begin{pmatrix} 0 & \mathbf{1}_{L-N} \end{pmatrix} W^{E,-} \begin{pmatrix} 0 \\ \mathbf{1}_{L-N} \end{pmatrix} \right)}. \end{aligned}$$

The statement then follows by the fact that the last factor is 1, since, by Theorem 2.4(e),

one has

$$\begin{aligned}
(0 \quad \mathbf{1}_{L-N}) W^{E,+} \begin{pmatrix} 0 \\ \mathbf{1}_{L-N} \end{pmatrix} &= (0 \quad \mathbf{1}_{L-N}) W \left(u_-^{\bar{E},-}, u_+^{E,+} \right) \begin{pmatrix} 0 \\ \mathbf{1}_{L-N} \end{pmatrix} \\
&= (0 \quad \mathbf{1}_{L-N}) W \left(u_-^{\bar{E},+}, u_+^{E,-} \right) \begin{pmatrix} 0 \\ \mathbf{1}_{L-N} \end{pmatrix} \\
&= (0 \quad \mathbf{1}_{L-N}) W^{E,-} \begin{pmatrix} 0 \\ \mathbf{1}_{L-N} \end{pmatrix}. \quad \square
\end{aligned}$$

Theorem 4.2.4. *We have*

$$\lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_{\sigma(H_0) + i\varepsilon} \frac{\det(S^E)'}{\det(S^E)} dE = L - \left(J_b + \frac{1}{2} J_h \right).$$

Proof. By Proposition 4.2.3 and the additivity of the logarithmic derivative, one immediately gets

$$\frac{\det(S^E)'}{\det(S^E)} = \frac{\det(W^{E,-})'}{\det(W^{E,-})} - \frac{\det(W^{E,+})'}{\det(W^{E,+})}.$$

The statement then follows immediately by Proposition 4.1.7 and the fact that, by [BS26c, TODO], one has $\sigma(H_0) = [-2a_1, 2a_1]$. $\square$

Acknowledgment: Miguel Ballesteros: [TODO]

Jonas Schober: Part of this work was conducted during my postdoctoral stay at the Instituto de Investigaciones en Matemáticas Aplicadas y en Sistemas at the National Autonomous University of Mexico in Mexico City and supported by the UNAM Postdoctoral Program (POSDOC). The latter part of the work was conducted at the Department of Mathematics at Brigham Young University, wherefore I would like to thank its sponsoring institution, The Church of Jesus Christ of Latter-day Saints together with all of its tithe-payers, who made this research possible.

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